

LESSON 2-10 • SESSION 2 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Find the Mean and MAD

Builds on Lesson 2-9 Session 1 • 6.SP.B.5c

TODAY'S LESSON

In Problem 3, Set W is symmetric, so you report its mean and its MAD.

YOU NEED TOFind the mean and the mean absolute deviation (MAD) of a data set. **You will use it in Practice Form A, problems 3 and 6.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Find** the mean: add, then divide by the count.
- 2 Find** each value's distance from the mean. Add them.
- 3 Divide** that total by the count. That is the MAD.

WATCH ME*the steps in action***Find the mean and MAD of 52, 54, 56, 58, 60.****Step 1** $280 \div 5 = 56$.**Step 2** Distances: 4, 2, 0, 2, 4. Total 12.**Step 3** $12 \div 5 = 2.4$.**Answer mean 56, MAD 2.4****TRY IT WITH HELP***fill in each step*

- 1.**
- Find the mean and MAD of 31, 33, 35, 37, 39.

Step 1 $175 \div 5 =$ _____**Step 2** Distances: 4, 2, 0, 2, 4. Total _____**Step 3** $12 \div 5 =$ _____

- 2.**
- Find the mean and MAD of 80, 82, 84, 86, 88.

Step 1 $420 \div 5 =$ _____**Step 2** Distances: 4, 2, 0, 2, 4. Total _____**Step 3** $12 \div 5 =$ _____**YOUR TURN***use the steps on your own*

- 3.**
- Find the mean and MAD.

a. 2, 4, 6, 8 _____**b.** 10, 10, 10, 10 _____**c.** 1, 5, 9, 13 _____**d.** 20, 24, 28, 32 _____

Choose Mean or Median

Builds on Lesson 2-10 Session 1 • 6.SP.B.5d

TODAY'S LESSON

In Problem 4, 400 is an outlier in the ticket sales, so the median is the center to report.

YOU NEED TO

Choose the better center: mean for symmetric data, median when there is an outlier or skew. **You will use it in Practice Form A, problems 1 and 5.**

STEPS TO FOLLOW

use these same steps on every problem

- 1 Look** for an outlier or a long tail.
- 2 Symmetric, no outlier:** report the mean (with the MAD).
- 3 Outlier or skew:** report the median (with the IQR).

WATCH ME

the steps in action

Which center fits 12, 13, 15, 17, 19, 21, 400?

- Step 1** 400 is an outlier.
Step 2 The data is not symmetric.
Step 3 Report the median, 17, with the IQR.

Answer median and IQR

TRY IT WITH HELP

fill in each step

- 4.** Which center fits 31, 33, 35, 37, 39?

- Step 1** Outlier? _____
Step 2 Shape: _____
Step 3 Report the _____ with the MAD.

- 5.** Which center fits 50, 54, 57, 59, 200?

- Step 1** Outlier? _____
Step 2 Shape: _____
Step 3 Report the _____ with the IQR.

YOUR TURN

use the steps on your own

- 6.** Write mean and MAD, or median and IQR.

- | | |
|--|---|
| a. 20, 22, 24, 26, 28 _____ | b. 3, 4, 5, 6, 900 _____ |
| c. 44, 46, 48, 50, 52, 54, 56 _____ | d. 80, 82, 84, 86, 88, 500 _____ |

READY CHECK

a problem like the one in today's lesson

- 7.** Fernanda reports a median of 70 for a skewed set. Which spread should go with it?

- A** the MAD **B** the IQR **C** the mean **D** no spread

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____